

Imaging the sub-moiré potential using an atomic single electron transistor

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Electrons in solids owe their properties to the periodic potential landscapes they experience. The advent of moiré lattices has revolutionized our ability to engineer such landscapes on nanometre scales, leading to numerous ground-breaking discoveries. Despite this progress, direct imaging of these electrostatic potential landscapes remains elusive. Here we introduce the atomic single electron transistor (SET), a new scanning probe that uses a single atomic defect in a van der Waals material as an ultrasensitive, high-resolution potential sensor. Built on the quantum twisting microscope (QTM) platform¹, this probe leverages the capability of the QTM to form a pristine, scannable two-dimensional interface between vdW heterostructures. Using the atomic SET, we present the first direct images of the electrostatic potential in a canonical moiré interface: graphene aligned to hexagonal boron nitride^{2–10}. The measured potential exhibits an approximate C_6 symmetry, minimal dependence on carrier density and a substantial amplitude of approximately 60 mV, even in the absence of carriers. Theory indicates that this symmetry arises from a delicate interplay of physical mechanisms with competing symmetries. The measured amplitude significantly exceeds theoretical predictions, suggesting that current understanding may be incomplete. With 1 nm spatial resolution and sensitivity to detect the potential of even a few millionths of an electron charge, the atomic SET enables ultrasensitive imaging of charge order and thermodynamic properties across a wide range of quantum phenomena, including symmetry-broken phases, quantum crystals, vortex charges and fractionalized quasiparticles.

The behaviour of electrons in a lattice is governed by the periodic potential of the host material. In naturally occurring materials, this periodicity is determined by the atomic length scale, making it extremely challenging to directly image the local electrostatic potential. Over the past decade, moiré engineering has emerged as a powerful approach to create tunable periodic potentials at van der Waals (vdW) interfaces, achieving length scales substantially larger than those of atomic lattices. A canonical example is the interface between aligned graphene and hexagonal boron nitride (G/hBN), in which their lattice mismatch produces a moiré superlattice that alters the electronic properties through its periodic potential. This moiré interface has enabled numerous discoveries, including the observation of the Hofstadter butterfly^{3–5} and Brown–Zak oscillations¹¹. More recently, when coupled with graphene multilayers, this aligned G/hBN interface has played a crucial role in stabilizing even more exotic phases, such as ferromagnetism in magic-angle twisted bilayer graphene^{12,13}, unconventional ferroelectricity in bilayer graphene^{14,15} and the fractional

quantum anomalous Hall effect (FQAHE) in rhombohedral pentalayer graphene¹⁶.

Despite its pivotal role, so far the G/hBN moiré potential has only been inferred indirectly from transport^{17,18} and optical^{19,20} measurements. Unlike in transition metal dichalcogenides (TMDs), whose band edges can serve as direct markers for potential variations²¹, mapping the moiré potential in a vdW interface such as G/hBN requires a new imaging technique that combines nanometre resolution with exceptional potential sensitivity. The most sensitive tool currently available for imaging electrostatic potentials is the scanning single electron transistor (SET)^{22–25}, which uses transport through a small island in the Coulomb blockade regime for detection. However, the spatial resolution of existing scanning SETs is constrained by their lithographic dimensions (>100 nm), preventing them from resolving potentials within a moiré unit cell. Although STM experiments with a graphene sensor layer^{26,27} have achieved intermediate spatial resolution, and recent advancements in AFM-based techniques have

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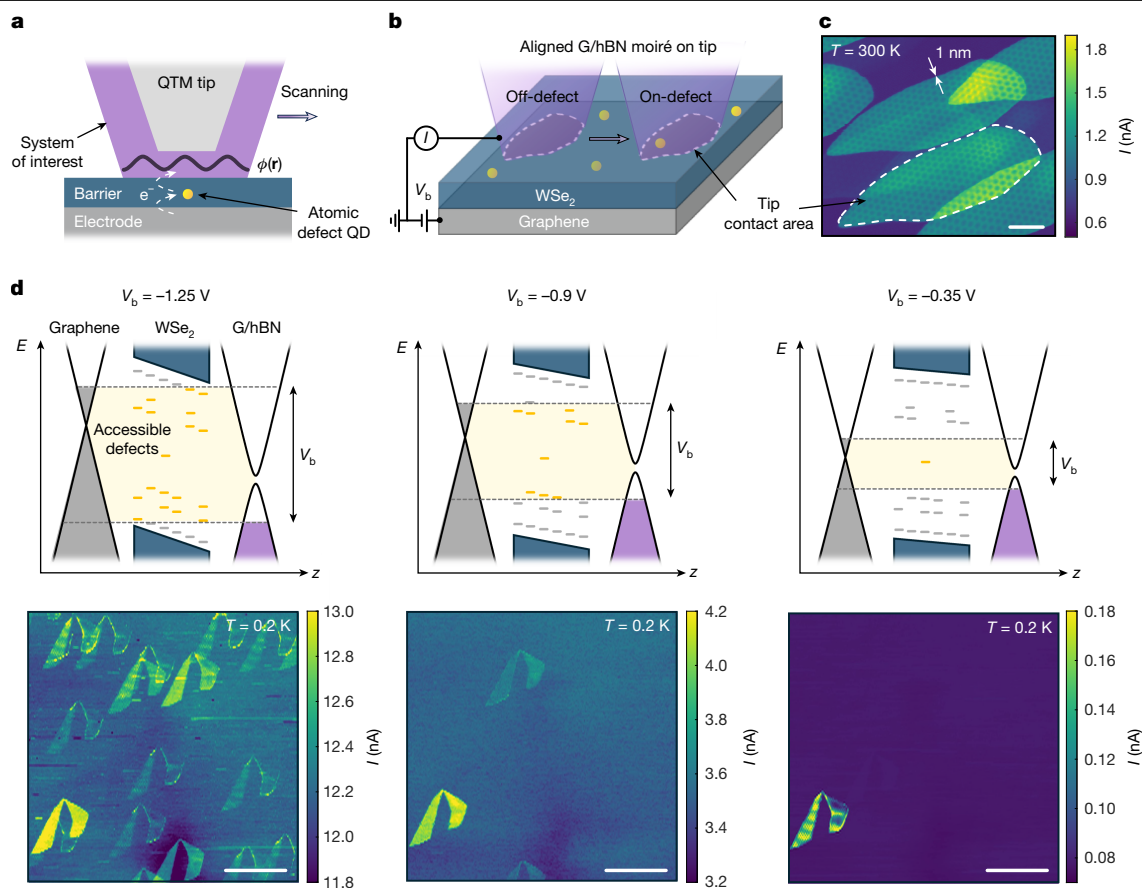


Fig. 1 | Atomic SET measurement principle and imaging of atomic defects.

a, Schematic of the atomic SET geometry: the system of interest (purple), positioned on a QTM tip, is scanned across a fixed atomic defect (yellow) embedded in an insulating barrier (blue) above a graphene electrode (grey). At low temperature, the defect functions as a quantum dot (QD) exhibiting single electron transport (white arrows). As the tip scans across this defect, the spatially varying electrostatic potential $\phi(r)$ of the system on tip gates the defect, and by monitoring its Coulomb blockade peak, we directly map this potential. **b**, Illustration of an experiment to image individual atomic defects in WSe_2 : a bias voltage (V_b) is applied to the bottom graphene electrode and the current (I) through the tip is measured while scanning. When the tip does not overlap a defect (off-defect), the current arises from momentum-conserving tunnelling processes. However, when the contact area of the tip (white dashed line) overlaps a defect (on-defect), an additional defect-assisted tunnelling pathway opens. **c**, Defect imaging experiment using a QTM tip comprising

aligned G/hBN layers. As the tip is scanned across trilayer WSe_2 on a graphene electrode at $V_b = -0.7$ V, the measured I map shows multiple replicas of the tip contact area, each produced by a different defect. Within each replica, the G/hBN moiré superlattice appears as periodic modulations of I and the sharp edges (arrows) demonstrate a spatial resolution of approximately 1 nm. **d**, Schematic energy diagrams (top) and corresponding current maps (bottom) measured with a different G/hBN tip at $T = 0.2$ K and at three biases: $V_b = -1.25$ V (left), -0.9 V (centre) and -0.35 V (right), all within the same scan window. The energy diagrams show the interface along the z -direction: a graphene source electrode (grey), defects (yellow and grey) in a bilayer WSe_2 barrier (blue) and a G/hBN moiré drain electrode (purple). As $|V_b|$ decreases, the range of transport-accessible defects (highlighted in yellow) is narrowed, resulting in fewer observed tip-shape replicas. At the lowest bias, only a single replica remains in the entire scan window. Scale bars, 50 nm (**c**); 400 nm (**d**).

led to high-resolution imaging of molecules^{28–30}, direct visualization of moiré potentials within vdW heterostructures remains an unmet challenge.

In this work, we develop the atomic SET, a new scanning probe that uses a single atomic defect^{31–36} as a scanning potential sensor, achieving 1 nm spatial resolution, two orders of magnitude better than existing SETs, and a potential sensitivity of $5 \mu\text{V Hz}^{-1/2}$. This remarkable sensitivity corresponds to detecting variations of a few parts per million of the potential produced by a single electron charge at the distance given by the spatial resolution (Supplementary Information section 17). Using this tool, we directly image the potential at the G/hBN moiré interface. Our measurements show that even at zero carrier density, the peak-to-peak potential amplitude is large (about 60 mV) and exhibits an approximate C_6 symmetry around the moiré centre. Although this symmetry can be explained by a subtle interplay of competing mechanisms, the magnitude of the measured potential is about twice that predicted by existing theory, highlighting

that our understanding of even this simple moiré interface remains incomplete.

Atomic defect sensing technique

The working principle of the scanning atomic SET is shown in Fig. 1a. A single atomic defect embedded in a thin TMD layer acts as a quantum dot, the energy level of which shifts with small changes to the local electrostatic potential $\phi(r)$. The system of interest is placed on the QTM tip, so scanning the tip across the stationary defect modulates the potential of the defect and, in turn, shifts its Coulomb blockade peak. By monitoring this shift, we directly image $\phi(r)$. This inverted geometry, in which the system of interest is on the tip, enables selecting an optimal defect from a large pool of natural defects within the flat TMD layer.

To locate suitable defects, we map the tunnelling current (I) at a fixed bias voltage (V_b) while scanning the tip across the TMD layer (Fig. 1b). When the tip does not overlap a defect, the current reflects only

background momentum-conserving elastic¹ and inelastic (phonon-assisted)³⁷ tunnelling. However, when it overlaps a defect, an additional tunnelling channel opens, allowing electrons to tunnel preferentially through the defect. This results in an increased I when the contact area of the tip (white dashed line) coincides with a defect.

Figure 1c shows a room-temperature measurement using a trilayer WSe₂ barrier. The scan shows several oblong shapes of increased I , each corresponding to a separate atomic defect imaging the contact area of the tip. Their identical spatial structure and increase in I suggest that these defects have the same chemical origin. In this experiment, the QTM tip consists of aligned G/hBN, forming a moiré superlattice^{2–5} that the defect imaging remarkably resolves even at room temperature. This measurement demonstrates extremely high spatial resolution, apparent from the sharpness of the tip image edges (about 1 nm; Supplementary Information section 6). Although we might expect that transferring a moiré heterostructure onto the QTM tip would result in significant strain that would be magnified by the moiré superlattice³⁸, this measurement (Fig. 3a) instead shows minimal moiré heterostrain, typically less than $\pm 0.3\%$ (Supplementary Information section 7).

Another important aspect of the defects is their energetics, which we investigate by imaging at different bias voltages at low temperature ($T = 0.2$ K). Figure 1d presents maps taken with another tip containing aligned G/hBN ($\lambda_m = 14$ nm) scanned over a bilayer WSe₂ barrier and graphene electrode. All subsequent data use this tip and are taken at this temperature. At high bias, the defect-assisted current is small compared with the Fowler–Nordheim tunnelling background through the WSe₂ conduction band³⁹. Nevertheless, numerous lungs-shaped replicas of the tip contact area appear, reflecting the large set of defects accessible at this energy. As the bias is reduced, progressively fewer defects contribute, and at the lowest bias, only a single low-energy defect remains within the scan window, with minimal residual background tunnelling. We use these relatively rare, low-energy defects for imaging: their sparse distribution ensures single-defect imaging, and their low energies allow us to operate near zero bias, avoiding hot-electron and phonon injection that could interfere with measuring the thermodynamic ground state of the system.

Quantum-dot-assisted spectroscopy

In the measurements so far, the defects served only as localized pathways for current. Now we aim to harness an individual defect as a fully functional quantum dot that can probe the local electrostatic potential and thermodynamic quantities at a specific position. This is accomplished by adding top and bottom gates to the QTM junction (Fig. 2a). In a prototypical quantum dot, applying a gate voltage V_{gate} linearly shifts the electrostatic potential of the quantum dot, producing a characteristic Coulomb diamond diagram (Fig. 2b). At zero bias, there is a specific V_{gate} in which the N and $N + 1$ charge states of the quantum dot are degenerate, permitting current to flow. At finite bias, the conduction window expands linearly with V_b , creating a diamond shape in the $V_{\text{gate}} - V_b$ plane.

In our experiment, the system of interest is situated between the gate and the defect, so the defect directly senses the local chemical potential of the system. As shown in Fig. 2c, the electrochemical (V), electrostatic (ϕ) and chemical (μ) potentials of the two graphene layers satisfy $\mu = V - \phi$ for each layer. Consequently, when both layers are grounded ($V_T = V_B = 0$), their local electrostatic potentials directly track their local chemical potentials. The defect experiences an electrostatic potential $\phi_D = \alpha\phi_T + (1 - \alpha)\phi_B$ (Supplementary Information sections 2 and 3), with $\alpha = z_B/(z_T + z_B)$ set by its relative distances to the top and bottom layers (z_T, z_B). The response of ϕ_D to a gate voltage V (top or bottom) is thus a weighted sum of the inverse electronic compressibilities $d\mu/dn$ of the top and bottom layers $\frac{d\phi_D}{dV} = \alpha \frac{d\mu_T}{dn_T} \frac{dn_T}{dV} + (1 - \alpha) \frac{d\mu_B}{dn_B} \frac{dn_B}{dV}$, where dn_T/dV and dn_B/dV are capacitive factors determined by the junction electrostatics (Supplementary Information section 2). Consequently, the

Coulomb diamond lines will curve in a way that reflects the Dirac-like $\mu(n)$ of both layers^{33,35,40–43} (Fig. 2d).

Before performing tunnelling measurements through a defect, we first establish the electrostatics of the QTM junction, which is controlled by the top gate (V_{TC}), the bottom gate (V_{BC}) and the bias (V_b) voltages. Figure 2e shows the tunnelling conductance, $dI/dV \equiv dI/dV_b$, measured off-defect as a function of V_{TC} and V_b . The two curved lines of reduced dI/dV correspond to the charge neutrality points of the top ($n_T = 0$) and bottom ($n_B = 0$) graphene layers, in agreement with the electrostatic model (dashed lines, Supplementary Information section 2). For $|V_b| < V_{\text{th}} \approx 65$ mV, there is a pronounced suppression of dI/dV . Detailed measurements (Supplementary Information section 9) indicate that this suppression is device-specific and arises from a large contact resistance that persists until the threshold bias V_{th} , then drops sharply. Therefore, for $|V_b| < V_{\text{th}}$, the bias primarily drops across the contact rather than across the QTM junction itself and the $n_T, n_B = 0$ lines remain vertical (essentially unaffected by bias). A similar measurement done in a perpendicular magnetic field of 5 T (Fig. 2f) shows suppressed dI/dV features associated with the Landau level gaps of the top ($v_T^{\text{LL}} = \pm 2, \pm 6$) and bottom ($v_B^{\text{LL}} = \pm 2$) layers, consistent with the electrostatic model. With additional measurements as a function of V_{BC} (Supplementary Information section 5), we fully establish how the three voltages control the carrier densities in the QTM junction.

We now measure tunnelling through a low-energy defect as a function of V_{TC} and V_b (Fig. 2g), observing an order of magnitude higher conductance. The conductance forms a curved Coulomb diamond: along its two branches, the defect level is resonant with either the top or bottom electrode (Supplementary Information section 2), whereas outside of this region, dI/dV is strongly suppressed, corresponding to fixed quantum dot charge (labelled as N and $N + 1$ for generality). For $|V_b| < V_{\text{th}}$, the Coulomb blockade peak is nearly vertical, consistent with the earlier off-defect observation that the bias drops primarily on the contact in this regime. For $|V_b| > V_{\text{th}}$, the curvature of the diamond edges reflects the $\mu(n)$ of the top and bottom layers. Specifically, two deflection points (arrows) correspond to the charge neutrality points of the two layers, in which the defect potential responds more strongly to gate voltage (that is, higher slope) due to the reduced compressibility of the top or bottom layer. In a similar measurement at $B = 5$ T (Fig. 2h), the Coulomb diamond edges show steps reflecting the Landau level gaps. Additional calibration experiments are presented in Supplementary Information section 16.

Imaging the G/hBN moiré potential

Having shown that a defect can measure $\mu(n)$ at a single point, we now turn to imaging potentials in real space. Figure 3a shows a high-resolution map of $I(x, y)$ measured through a single low-energy defect at $V_b = -0.35$ V, revealing the detailed moiré structure of the tip. To extract the electrostatic potential, we monitor how the zero-bias Coulomb peak shifts as the tip is scanned across the defect. Figure 3b shows the zero-bias $dI/dV(x, V_{\text{TC}})$ measured along the white dashed line in Fig. 3a. A narrow Coulomb blockade peak is observed at each x , whose gate-voltage position $V_{\text{TC}}^{\text{peak}}(x)$ oscillates with the moiré periodicity, following $V_{\text{TC}}^{\text{peak}}(x) = c \cdot \phi(x) + \text{const}$ (c is determined by the junction electrostatics; Supplementary Information sections 2 and 3). This scan, therefore, directly tracks the moiré electrostatic potential (Fig. 3b, right axis).

We can now apply this technique to obtain a full 2D map of the moiré potential. Figure 3c shows the zero-bias dI/dV measured as a function of x and y (Fig. 3a, dotted box) and at several values of V_{TC} . At the lowest V_{TC} , dI/dV is practically zero at every (x, y) position. With increased V_{TC} , conductance appears along rings, which repeat in the (x, y) plane with the moiré periodicity. These rings correspond to equipotential lines within the moiré unit cell. Further increasing V_{TC} increases the radius of the rings until they merge and disappear. From a full

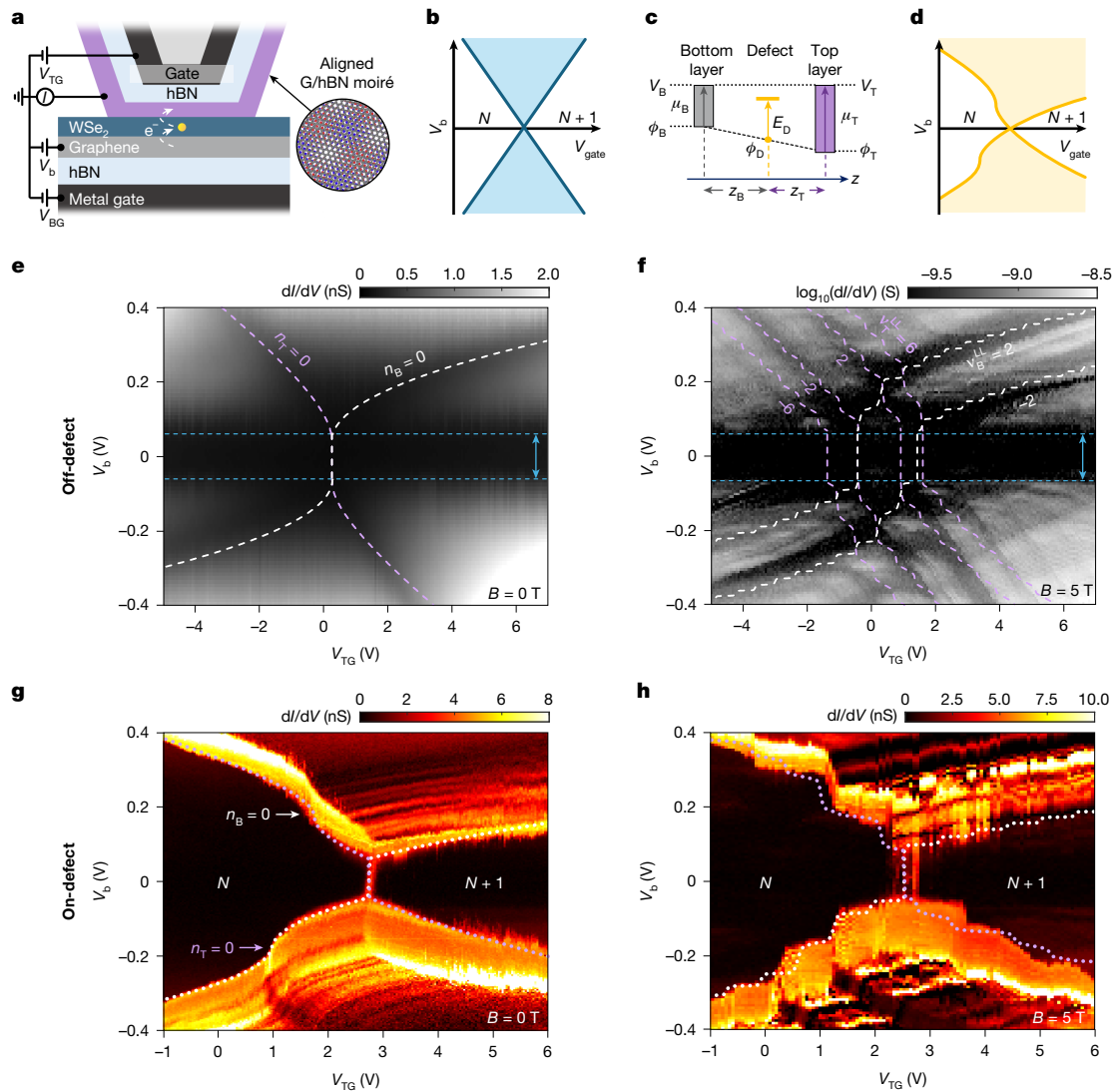


Fig. 2 | Measuring local chemical potential using an atomic defect quantum dot. **a**, Cross-section of the QTM junction: the tip comprises an aligned G/hBN moiré interface, an hBN layer and a graphite top gate. The sensor device contains a defect-bearing bilayer WSe₂ barrier, a graphene electrode, an hBN layer and a metal bottom gate. **b**, Prototypical quantum dot Coulomb diamond conductance diagram versus gate voltage (V_{gate}) and source-drain bias (V_b), showing a central conduction region (blue) and Coulomb-blockaded regions with fixed charge states (labelled N and $N+1$). **c**, Electrostatic relations in the junction: for each graphene layer, $\mu = V - \phi$. The defect potential is $\phi_D = \alpha\phi_T + (1-\alpha)\phi_B$ with $\alpha = z_B/(z_T + z_B)$, where z_B and z_T are the distances of the defect from the bottom and top layers, respectively. The defect energy E_D is referenced to ϕ_D . **d**, Because the system of interest lies between the gate and the defect, gate voltages are screened by the electronic compressibility of the system. This leads to curvature in the Coulomb diamond lines, reflecting the density-dependent $\mu(n)$ of the two graphene electrodes. **e**, Off-defect differential conductance (dI/dV)

versus top gate voltage (V_{TG}) and V_b at $T = 0.2$ K and $B = 0$ T. Curves of reduced dI/dV correspond to the charge neutrality points (CNPs, $n = 0$) of the top (purple dashed) and bottom (white dashed) graphene layers. Additional suppression for $|V_b| < V_{\text{th}} \approx 65$ mV (horizontal cyan dashed lines) arises from nonlinear contact resistance (see text). **f**, Off-defect measurement at $B_{\perp} = 5$ T showing reduced dI/dV along Landau level gaps in the top ($v_T^{\text{LL}} = \pm 2, \pm 6$, purple) and bottom ($v_B^{\text{LL}} = \pm 2$, white) graphene layers. **g**, On-defect dI/dV versus V_{TG} and V_b at $B = 0$ T, exhibiting a Coulomb diamond diagram with curved boundaries that separate blocked regions with near-zero dI/dV and fixed quantum dot charge (N and $N+1$) from a high-conductance region. Arrows mark the deflection points at the CNPs of the top G/hBN (purple) and bottom graphene (white) layers. **h**, On-defect measurement at $B_{\perp} = 5$ T, where the diamond edges exhibit step-like features due to Landau level gaps in both electrodes. White and purple dashed lines in **e–h** are fits to the electrostatic model (Supplementary Information sections 2 and 5).

three-dimensional map of $dI/dV(x, y, V_{\text{TG}})$ (Supplementary Video 1), we extract $V_{\text{TG}}^{\text{peak}}(x, y)$ (Supplementary Information section 4), and from the relation above, directly determine the 2D moiré potential.

Figure 3d shows the 2D moiré potential $\phi(x, y)$ across a single moiré unit cell at three fillings, $\nu = 0 \pm 0.15, 1.3 \pm 0.2$ and 4 ± 1 (with $\nu = 1$ corresponding to one electron per moiré cell). Several key features emerge. First, the moiré potential amplitude is large, ranging from 52 mV to 62 mV (Supplementary Information section 18). Second, this amplitude varies only minimally (about 10%) with moiré filling. The maps also show three distinct high-symmetry points: a central maximum (red) and two minima

separated by 60° (grey and blue). The difference between these minima, $\Delta\phi_{\text{minima}} \approx 4$ mV, is a small fraction of the overall scale. This indicates that although a minor C_3 component is present, consistent with the underlying symmetry of the moiré lattice, the overall symmetry is close to C_6 .

Comparison with theory

To interpret our observations, we consider various mechanisms that have been theoretically proposed to induce potential. Following ref. 10, the moiré Hamiltonian can be decomposed into three terms:

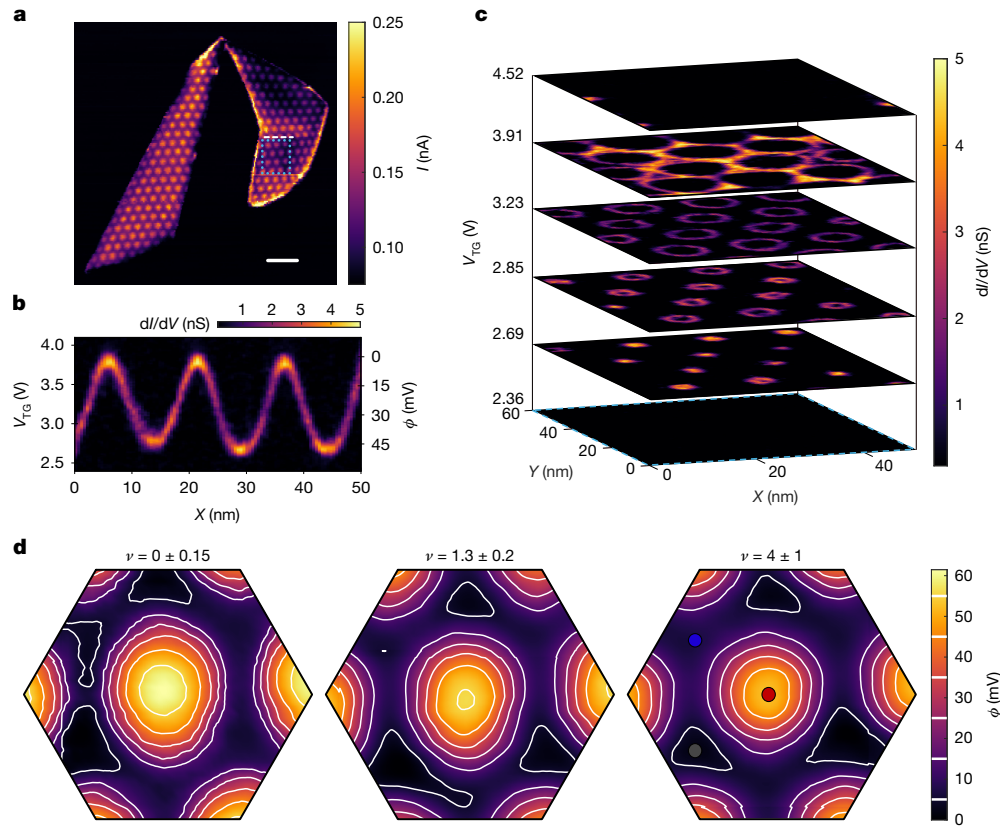


Fig. 3 | Imaging the moiré potential of aligned G/hBN using the atomic SET.

a, High-resolution current map $I(x, y)$ at $V_b = -0.35$ V and $T = 0.2$ K, showing the shape of the contact area of the G/hBN tip and the detailed moiré structure within. **b**, Zero-bias dI/dV versus x and V_{TG} , measured at filling $\nu = 3.2 \pm 1$. The narrow Coulomb-blockade peak oscillates with the moiré periodicity; its gate-voltage position $V_{TG}^{\text{peak}}(x)$ is converted to the electrostatic potential at the moiré interface, $\phi(x)$ (right y-axis), using the junction electrostatics (Supplementary Information sections 2 and 3). This measurement cuts through the high-symmetry sites of the moiré superlattice with the highest $\phi(r)$ (white dashed line in **a**). **c**, Extension of **b** to two spatial dimensions (blue dotted region in **a**), showing dI/dV measured versus x, y and V_{TG} for a few slices of constant V_{TG} . At the lowest V_{TG} , dI/dV is practically zero everywhere. As V_{TG} increases, rings of conductance appear and repeat with the moiré periodicity, corresponding to

equipotential lines that expand and merge with increasing V_{TG} . From the full three-dimensional dataset (Supplementary Video 1), we extract $V_{TG}^{\text{peak}}(x, y)$ (Supplementary Information section 4) and from the junction electrostatics directly determine the moiré potential, $\phi(x, y)$. **d**, $\phi(x, y)$ of aligned G/hBN, measured at fillings $\nu = 0 \pm 0.15$ (left), $\nu = 1.3 \pm 0.2$ (centre) and $\nu = 4 \pm 1$ (right). These maps are obtained by averaging over several moiré sites (uncertainties in ν arise from the V_{TG} adjustments used to meet the quantum dot resonance condition; Supplementary Information sections 4 and 13). We set $\phi = 0$ at the potential minimum. The centre and right maps use defect D1 at zero bias; the left map uses defect D2 at $V_b = -0.21$ V. The potential exhibits an approximate C_6 symmetry, changes minimally with carrier density (about 10%) and has a substantial amplitude even at zero carrier density. High-symmetry stacking sites are indicated by red, blue and grey dots. Scale bar, 50 nm (**a**).

an effective pseudoelectric potential (H_0), a pseudomagnetic field (H_{xy}) and a local mass term (H_z), each associated with a corresponding sublattice Pauli matrix (Supplementary Information section 13). The dominant H_0 term arises from two effects: (1) the spatially varying stacking potential generated by changes in the local G/hBN alignment within the moiré cell and (2) the in-plane relaxation of the graphene lattice, which stretches and compresses the C–C bonds, leading to local variations in the Dirac point energy relative to the vacuum level and captured by the deformation potential^{9,44}.

Figure 4 shows the predicted stacking (Fig. 4a) and deformation (Fig. 4b) pseudopotential terms comprising H_0 , along with the pseudomagnetic field H_{xy} (Fig. 4c), with the three high-symmetry CB (carbon above boron), CN (carbon above nitrogen) and AA (carbon above both boron and nitrogen) stacking sites marked. Generally, a pseudoelectric potential may reflect energy changes that are not electrostatic in nature. However, theory⁴⁴ and density functional theory calculations^{45,46} suggest that the pseudopotentials in Fig. 4a,b are directly accounted for by charge polarization perpendicular to the layers and therefore manifest as real electrostatic potentials.

Graphene carriers redistribute to screen these pseudopotentials, producing the self-consistent electrostatic potential measured by our

detector. We model this using self-consistent Hartree calculations (Supplementary Information section 13). The resulting screened potentials from the components in Fig. 4a–c are shown in Fig. 4d–f. Notably, screening preserves the shape of the pseudoelectric potentials but reduces their magnitude by approximately 2.2, consistent with the predicted random-phase-approximation dielectric constant $\epsilon = 1 + \pi\alpha/2 \approx 2.0$, where $\alpha = \frac{e^2}{4\pi\kappa\epsilon_0\hbar v_F}$ is the fine-structure constant of graphene, v_F is its Fermi velocity and $\kappa = 3.5$ is the hBN dielectric constant⁴⁷. Furthermore, we find that electronic screening converts the pseudomagnetic field H_{xy} into an electrostatic potential (Fig. 4f). This potential is small compared with the other two, scales linearly with ν , and becomes identically zero at $\nu = 0$ (same for H_z ; Supplementary Information section 13). As our experiments show only a minor ν dependence, we omit the H_{xy} and H_z terms in further discussions.

Both leading potential terms (Fig. 4d,e) exhibit a clear C_3 symmetry around the central CN site, in contrast to the approximate C_6 symmetry observed experimentally. However, examining the minima at the CB and AA sites shows that these C_3 symmetries are inverted—for the first term $\phi_{AA} > \phi_{CB}$ and for the second term $\phi_{CB} > \phi_{AA}$. The two terms compensate each other to form an almost C_6 -symmetric total potential with a pronounced central peak (Fig. 4g). The resulting total self-consistent

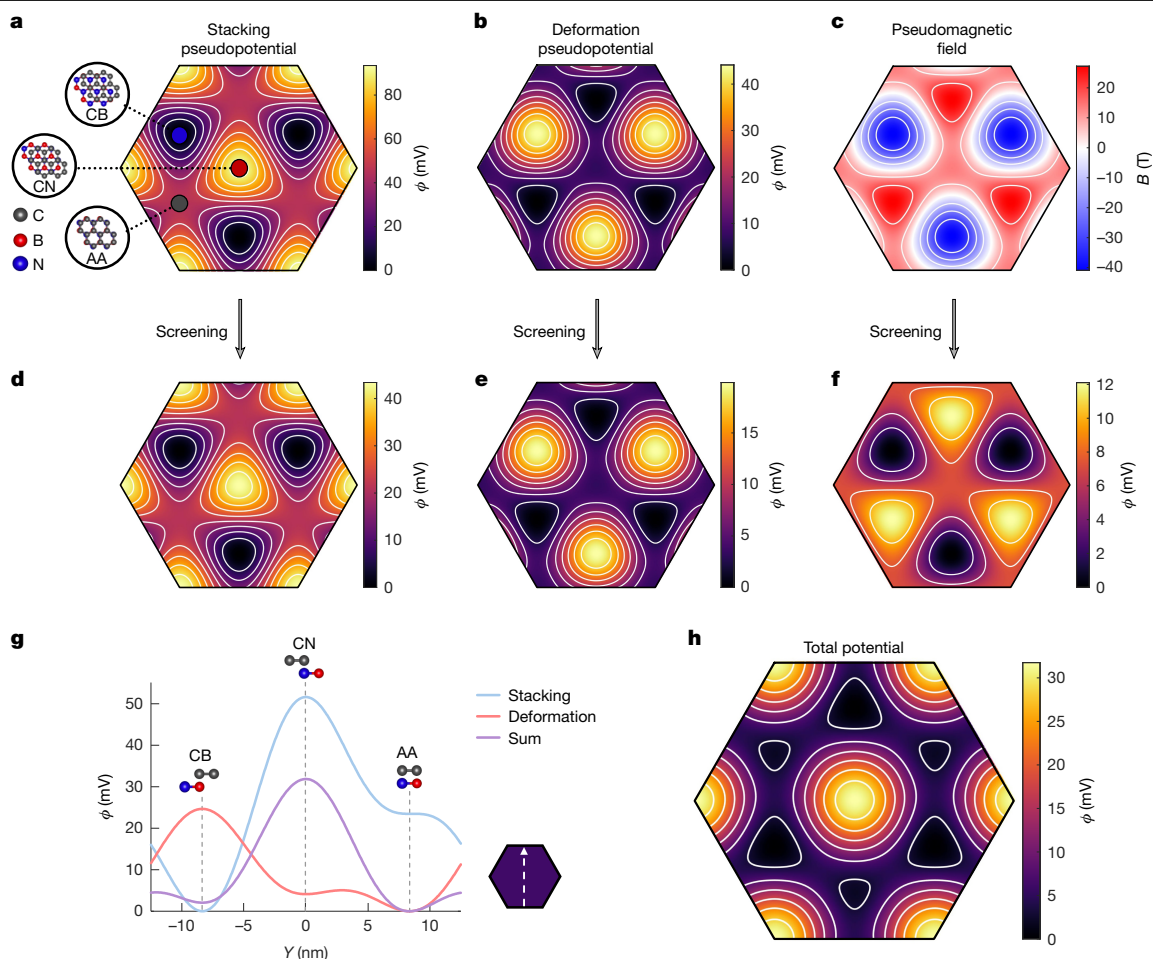


Fig. 4 | Theoretical breakdown of the various physical mechanisms contributing to the moiré potential in aligned G/hBN. **a**, Stacking pseudopotential due to the relative stacking of G and hBN, which varies within the moiré unit cell. The three high-symmetry points corresponding to local CB, CN and AA stacking are marked. **b**, Deformation pseudopotential due to atomic relaxation within the graphene layer. The terms in **a** and **b** both appear in the H_0 part of the Bloch Hamiltonian, corresponding to the identity matrix in the sublattice basis. **c**, Magnitude of the pseudomagnetic field, which appears in the H_{xy} part of the Hamiltonian, corresponding to the σ_x and σ_y Pauli matrices in the sublattice basis. **d–f**, Self-consistent electrostatic potentials obtained after considering the screening by the graphene carriers, calculated by including a self-consistent Hartree potential response using a carrier density corresponding

to $\nu = 4$. All terms show strong C_3 symmetry around the moiré centre, in contrast to the C_6 symmetry observed in the experiments. **g**, Self-consistent stacking and deformation potentials are plotted along a linecut through the moiré centre (dashed white line, bottom inset). The CB, CN and AA high-symmetry points are labelled. Visibly, each of the two terms (blue, pink) shows a strong C_3 symmetry. However, owing to cancelling contributions, their sum (purple) exhibits an approximate C_6 symmetry, with only a small difference between the potential minima at the CB and AA stacking sites. **h**, Total self-consistent potential calculated for $\nu = 0$. This potential resembles the experiment in terms of the approximate C_6 symmetry, but its magnitude is half of that measured experimentally.

potential (Fig. 4h) strongly resembles the experimental result, with one notable exception—the experimental potential scale is double the theoretical prediction. One explanation might be that theory underestimates strain in the moiré interface²¹. However, increasing strain alone would yield a more C_3 -symmetric potential, contrasting our observations. This large discrepancy demonstrates that, despite G/hBN being one of the most relevant and extensively studied moiré interfaces, there are substantial gaps in its theoretical understanding, which has direct consequences for recent experiments that use this interface to design new states of matter (for example, FQAHE in moiré pentalayer graphene).

Finally, we show the dependence of the moiré potential on the distance from the moiré interface. Extended Data Fig. 1 presents potential traces measured by two defects, located at approximately 0.8 nm (D2) and 1.5 nm (D1) from the interface (Supplementary Information section 3). The measured potential decays rapidly, even over these small distances. This significant drop suggests that if the detector were at a moiré distance ($h = \lambda_m$) away from the interface, it would have detected

only $e^{-\frac{4\pi}{\sqrt{3}}\left(\frac{h}{\lambda_m}\right)\sqrt{\frac{eV}{\epsilon_1}}} \approx 10^{-4}$ of the potential, underscoring the importance of our atomic SET operating at extremely close standoff distances. At the same time, these measurements also demonstrate that in thin flakes, such as pentalayer graphene, electrons can still experience a significant moiré potential (tens of mV) even in the farthest graphene layer.

The atomic SET scanning probe technique demonstrated here has a combination of features that are extremely powerful for studying a wide range of quantum materials. Its QTM geometry allows it to scan within the pristine interfaces of a variety of vdW materials. Similar to existing SETs, this technique will allow quantitative measurements of thermodynamic properties such as the electronic compressibility^{24,25,48} and entropy⁴⁹, but now with two orders of magnitude improved spatial resolution—below the Fermi wavelength, magnetic length and moiré scales of many systems. This advance extends this powerful imaging method to a much broader class of physical phenomena occurring on small scales such as Wigner crystals, topological edge states, vortex charges, symmetry-broken phases and fractionally charged quasiparticles.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-10085-z>.

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Data availability

The data shown in this paper are provided at zenodo.org/records/17689195. Additional data that support the plots and other analysis in this work are available from the corresponding author upon request.

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Author contributions D.R.K., U.Z. and S.I. designed the experiment with the assistance of J.B., A.I. and J.X. in the early stages of the experiment. U.Z. built the millikelvin scanning microscope. D.R.K., U.Z., A.K., Y.Z. and M.S. fabricated the devices and performed the experiments. D.R.K., U.Z., A.K. and S.I. analysed the data. M.M.A.E., L.P. and S.A. wrote the theoretical model. K.W. and T.T. supplied the hBN crystals. D.R.K., U.Z. and S.I. wrote the paper with input from other authors.

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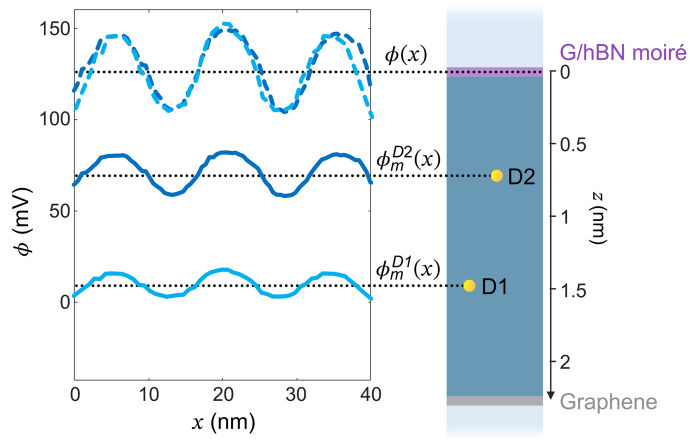
Additional information

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Extended Data Fig. 1 | Measured decay of the moiré potential with height. Linecuts along high-symmetry points of the moiré potential, ϕ_m^D , measured as a function of position, x , using two defects (D1 and D2). These defects are located at different heights from the G/hBN interface (1.5 nm and 0.8 nm, respectively) and measured at moiré fillings $\nu = 4 \pm 1$ and $\nu = 3.2 \pm 1$. We also plot the potential at the moiré interface ($\phi(x)$, dashed lines) deduced using the calibrated junction electrostatics (Supplementary Information sections 2 and 3) from the measurements with both defects. The potentials are offset along the y-axis for clarity. These measurements demonstrate that even at such small heights, the decay of the moiré potential amplitude is substantial (~60% measured decay from 0.8 nm to 1.5 nm).